

Youna Jang

(+82)10-4538-5854

ynjjang01@gmail.com

<https://squirtttt.github.io/>

Research Interest

Neural Architecture Search

Medical Image Analysis

Vision-Language Model

Educations

Mar 2024 – **M.S. in Artificial Intelligence Applications (4.42/4.5)**
Feb 2026 *Neural Architecture Search for Medical Image Segmentation*
Kwangwoon University, Seoul, Republic of Korea
Mar 2020 – **B.S. in System Software (3.79/4.5)**
Feb 2024 Kwangwoon University, Seoul, Republic of Korea

Publications

Jang, Y.N., Lee, S.M., **HB-NAS: Hierarchical Bayesian Neural Architecture Search**
(On working)

Jang, Y.N., Lee, S.M., **ZAMS: Zero-shot Adapter Architecture Search for Segmentation Foundation Models**
(Under review at CIKM 2026)

Jang, Y.N., Lee, E.H., Go, H.J., Park, J.Y., Lee, S.M., **S3DM: Semi-supervised Signed Distance Maps for Boundary-Aware Tissue Segmentation**
(Peer review at BMC Medical Imaging)

Noh, Y.J., Baek, S.J., Jang, H.Y., Jang, Y.N., Lee, S.M., Lee, S.G., **OOD-Based Object Detection for Forklift Remote Loading Assistance System**
Journal of the Korea Society of Electronic Commerce 29.4 (2024):185-202

Research Experiences

- Conducted research on Neural Architecture Search (NAS) and efficient architecture for medical image analysis.
- Developed uncertainty-aware Bayesian optimization and foundation model adaptation methods for robust AI systems.
- Applied machine learning and Vision-LLM techniques to industrial AI applications, including autonomous equipment control and production optimization.

Conferences

Apr 2026 Jang, Y.N., Lee, S.M., **Efficient SAM Adaptation through Zero-shot Architecture Search**
International Conference on Industrial Engineering and Applications, Kyoto, Japan, 2026 (April 09-12)

Aug 2025 Jang, Y.N., Lee, S.M., **Efficient Zero-shot Neural Architecture Search for SAM-Adapter Architecture** (poster)
Korea Data Mining Society, Pyeongchang, Republic of Korea, 2025

Jun 2025 Jang, Y.N., Lee, S.M., **Zero-shot based SAM Adapter Architecture Search**
Korea Institute of Industrial Engineers, Jeju, Republic of Korea, 2025 (June 18-22)

Nov 2024 Jang, Y.N., Lee, S.M., **S3D-NAS: Self-Distilled Dirichlet Distribution Neural Architecture Search in Medical Image Segmentation** (poster)
Korea Data Mining Society, Gyeongju, Republic of Korea, 2024 (November 22-23)

Nov 2024 Jang, Y.N., Go, H.J., Park, J.Y., Lee, S.M., **Signed Distance Map based Semi-supervised Semantic Segmentation in Histopathology** (poster)
Korea Data Mining Society, Gyeongju, Republic of Korea, 2024 (November 22-23)

May 2024 Jang, Y.N., Lee, S.M., **Signed Distance Map-oriented Contrastive Learning for Semantic Segmentation in Magnetic Resonance Imaging**

Awards & Honors

Apr 2026	Best Oral Presentation
	Efficient SAM Adaptation through Zero-shot Architecture Search <i>International Conference on Industrial Engineering and Applications, Kyoto, Japan, 2026</i>
Aug 2025	Best Paper Award
	Efficient Zero-shot Neural Architecture Search for SAM-Adapter Architecture <i>Korea Data Mining Society, Pyeongchang, Republic of Korea, 2025</i>
Nov 2024	Best Paper Award
	S3D-NAS: Self-Distilled Dirichlet Distribution Neural Architecture Search in Medical Image Segmentation (Poster) <i>Korea Data Mining Society, Gyeongju, Republic of Korea, 2024</i>

Projects

Jan 2026 ~ Present	Development of a Vision-LLM based Autonomous Equipment Control AI Agent
	Samsung Electronics – Development of a Vision-LLM-based agentic AI to derive optimized reasoning and action sequences for equipment control system
Jan 2026 ~ Present	Efficiency Optimization for AI-driven Production Facility Layout Design Automation
	Samsung Electronics – Facility layout optimization and KPI development using Bayesian Optimization and Meta-heuristic methods
Oct 2025 ~ Jan 2025	Development of Finance-specialized customer service
	Semyung Soft – Developed an analysis system for finance specialized customer dialogue histories after QLoRA-based fine-tuning
Nov 2024 ~ Dec 2025	Development of Hotel Concierge Chatbot
	Stayneo – Developed a concierge-specialized chatbot service using LLMs tailored for hotel guest support
Mar 2024 ~ Oct 2025	Development of Remote Operation and Assistance System for Onboard Loading Equipment
	Korea Railroad Research Institutes – Built a real-time assistance system using LiDAR and depth cameras, contributing to emergency braking for worksite safety
Mar 2024 ~ Dec 2024	Quarry Site Operation Optimization using AnyLogic Simulation
	HD Hyundai Site Solution – Conducted system operation data analysis to support simulation-based optimization of quarry site operations

Teaching Experiences

Dec 2025	Data Analytics and AI Education: Python, Data Analytics, and Machine Learning
	Instructor – Samsung Electronics “Data Analysis Boot Camp”
Jan 2025 – Jun 2025	Post-project Consulting: Time Series Forecasting and Recommendation System
	Mentor – Samsung Electronics “Data Analysis Boot Camp”
Dec 2024	Data Analytics and AI Education: Python, Data Analytics, and Machine Learning
	Instructor – Samsung Electronics “Data Analysis Boot Camp”
Sep 2020 – Feb 2024	Coding Education for K-12 Students: Python, Arduino, and Basic Programming
	Volunteer Instructor – Sowon club (Coding Outreach Program)

References

Professor Sangmin Lee

M.S. Supervisor

20, Gwangun-ro, Nowon-gu, Seoul, Republic of Korea

smlee5679@gmail.com